COURSE TITLE : **PRODUCTION ENGINEERING – II**
COURSE CODE : **4113**
COURSE CATEGORY : **B**
SEMESTER : **4**
PERIODS PER WEEK : **4**
PERIODS PER SEMESTER : **60**
CREDITS : **4**

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Milling ,Indexing	15
2	Cutting Fluids Grinding	15
3	Jig Boring, Jig Grinding, Horning, Laping and Super finishing Broaching,	15
4	Jigs and Fixtures	15
	TOTAL	**60**

COURSE OUTCOMES

After completion of the study the students will be able to: -

1. Understand the working of milling machine.
2. Know different indexing methods
3. Understand the types of cutting fluids and their application
4. Comprehend different operations carried out in various types of grinding machines
5. Know the methods of Jig boring and Jig grinding
6. Comprehend the working and use of broaching machine
7. Understand the basic principles and use of Jigs and Fixtures

SPECIFIC OUTCOMES
MODULE I

1.1.0 Understand the working of milling machine carrying out different types of work

1.1.1. Describe general uses of milling machine
1.1.2. Identify part of milling machine and there function
1.1.3. Identify different types of milling machine Viz plain, universal and vertical
1.1.4. Identify nomenclature of milling cutters
1.1.5. Specify cutting materials and milling cutters
1.1.6. Suggest cutter holding devices

1.1.7. Select suitable work holding devices, speed, feed and depth of cut for various operation
1.1.8. Describe different milling methods Viz up milling down milling
1.1.9. Discuss different milling operations such as plain milling, keyway milling, T slot milling and gang milling

1.2.0 Know different indexing methods

1.2.1 Specify a milling machine

MODULE II

2.1.0 Analyse the different types of cutting fluids and their application

2.1.1 State the functions of cutting fluids
2.1.2 List the classification of cutting fluids
2.1.3 State the application of cutting fluids during machining operation
2.1.4 Describe the effect of cutting fluids on tool life
2.1.5 Select the appropriate cutting fluid for a given applications

2.2.0 Comprehend different operations carried out in various types of grinding machines

2.2.1. Classify the types of grinding machine
2.2.2 Describe different special purpose of grinders Viz cylindrical grinder, centreless grinder
2.2.2 Describe arrangement of work between centers setting up of grinding wheels, method of supporting work in these grinding machine
2.2.3 Compare different types of grinding
2.2.4 Describe surface grinders of different types
2.2.5 Identify different types of work holding devices in surface grinder
2.2.6 Discuss a tool and cutter grinder
2.2.7 Illustrate the methods of balancing the wheel, truing and dressing
2.2.8 Select coolants for grinding
2.2.9 Select speed, feed and depth of cut for different materials

MODULE III

3.1.0 Know the methods of Jig boring and Jig grinding

3..1.1 Describe different types of Jig boring machines
3.1.2 Describe Jig grinding machine of different types
3.1.3 Select suitable grinding wheels and work holding device
3.1.4 State the basic features of honing, laping and super finishing

3.2.0 Comprehend the working and use of broaching machine

3.2.1 Identify different parts of a broaching tool
3.2.2 State different copying machine

MODULE IV

4.1.0 Understand the basic principles and use of Jigs and Fixtures

4.1.1 Differentiate between Jigs and Fixtures
4.1.2 Discuss the principles of locator
4.1.3 Discuss the clamping devices
4.1.4 Identify different clamping devices employed in Jigs and Fixtures
4.1.5 List out different types of jigs & fixture
4.1.6 Identify different types of bushes in jig
4.1.7 Identify the materials used for jigs and fixtures
4.1.8 State application of fixtures

CONTENT OUTLINES

MODULE I

Milling

General use of milling machines – parts of milling machine and their functions – worktable, saddle, over arm, spindle, column and knee. Milling machine specification
Types of milling machines – plain, universal and vertical milling cutters
Cutter holding devices – arbor, collet and adaptors setting up of a work – work holding devices – alignment – speed, feed and depth of cut for various materials – milling operations – conventional milling, climb milling, plain milling, key and key ways, gang milling and T-slot milling.

Indexing

Indexing head – functions – indexing calculations – direct indexing, simple indexing and differential indexing.

MODULE II

Cutting fluids

Functions of cutting fluids – classification – basic actions of cutting fluids – application of cutting fluids during machining operation – effect of cutting fluids on tool life – selection of cutting fluids – Quantity of cutting fluids circulated to remove heat during machining process

Grinding

Grinding machines – classification – cylindrical grinders, surface grinders.
Special purpose grinders – cylindrical grinders – arrangement of work between centers, set up of grinding wheel, methods of supporting work between in a centreless grinder. Action of grinding wheel, action of regulation wheel, feed of work piece.
Advantages of centreless grinding, constructional features of internal grinder

Surface grinder – methods of holding work, methods of mounting the wheel accessories – magnetic chucks – construction, action, permanent magnet chuck, direct current chuck methods of balancing the wheel, truing and dressing, coolants used, values of speed, feed, and depth of cut for materials such as high carbon steel and alloy steel.

MODULE III

Jig boring and Jig grinding

Introduction – definition – need for greater accuracy on jigs, button boring, jig boring on a vertical milling machine.
Types of jig boring machines – open front type and cross rail type machines

Basic features of finishing by honing,laping and super finishing

Broaching

Definition – purpose – broaching tools – parts of the tool

MODULE IV

Jigs and fixtures

Jigs and fixtures – introduction – definition of jigs and fixtures. Difference between jig and fixture
Locating principles – confinement and clamping devices – clamping principles – types of clamps – screw clamping devices cam clamps – toggle clamps, strap clamps, cams, wedges, latches, floating clamps, rack and pinion clamps, vacuum clamps, pneumatic and hydraulic clamps
Jigs: Type of jigs – open type, box type, plate type, swinging plate type – materials for jigs and fixtures and their heat treatment
Jig bushes – types – methods of making
Fixtures: application and types – lathe fixture, milling fixture, grinding fixture

REFERENCE BOOKS

1. A text book of Production Engineering – P.C. Sharma
2. A text book of Production Technology – P.C. Sharma
3. Jigs and Fixtures – Joshi
4. Jigs and Fixtures – Franklin.D.Jones
5. Robotics for Engineers – Yoram Koren
6. Production Technology – HMT
7. Production Technology – R.K. Jain & S.C. Gupta
8. Numerical Machine Tools – S.J Martin
9. N.C. Machines and CAM – Pressman Williams
10. Tool Design – Donaldson
11. Work Shop Technology –Vol.II -S.K.Hajra Chandhuri
12. Machine tools and tool design - P.C Sharma (SChnad)